MATHEMATICS
Introduction to Modern Algebra II
Quiz 2.

1) Let F be a field. Prove that $F[x]$ is a PID, in other words show that any ideal of $F[x]$ is principal (i.e. generated by only one element). Give an example of a ring with an ideal not generated by one element (i.e. a ring which is not a PID).

Proof. Let $I \subset F[x]$ be an ideal. We may assume $I \neq (0)$. By the WOP, we may pick $0 \neq f(x) \in I$ of minimal degree. For any $g(x) \in I$, we may write $g(x) = f(x)q(x) + r(x)$ where $\deg r(x) < \deg f(x)$. Since $f(x), g(x) \in I$, we have $r(x) = g(x) - f(x)q(x) \in I$. Since $\deg r(x) < \deg f(x)$, we have $r(x) = 0$. But then $g(x) = f(x)q(x) \in I$ as required.

$(2, x) \subset \mathbb{Z}[x]$ and $(x, y) \subset \mathbb{R}[x, y]$ are not principal ideals. $\square$

2) Let F be a field. What are the invertible elements of $F[x]$ (prove your answer).

Proof. The invertible elements of $F[x]$ are given by F^*. It is clear that every element of F^* is invertible. Suppose now that $f(x) \in F[x]$ is invertible. Then $f(x)g(x) = 1$ for some $g(x) \in F[x]$. But then $\deg f(x) + \deg g(x) = \deg 1 = 0$ and so $\deg f(x) = \deg g(x) = 0$. But then $f(x)$ is a non-zero constant i.e. $f(x) \in F^*$. $\square$